Examiner reports

Q1.

Part (a) was successfully attempted by well over half of the students. However, the value of g to be used in this question was 9.81 and a significant number of students gave their answer to the wrong degree of accuracy, usually costing them one mark. This is the only place in the paper where this was penalised.

There were many inefficient methods seen for this question. There seemed an over reliance on set routines rather than choosing the best equation to suit the known information. For instance, several students doubled the time and set the vertical displacement to zero. This works and the method received full credit if done correctly, but these students often showed a lack of intuition or insight.

In part (b) most students made some progress, but there were several error prone, inefficient methods seen. A significant number of students unnecessarily split the motion of the projectile into two stages.

In this question, correct answers only received full credit if they included the correct units, which was not always the case.

Q3.

There were very mixed responses to this question. Some students worked through this very easily, but others found it difficult to get started and produced very little meaningful work. A very common issue was mixing up aspects of the horizontal and vertical motion. For example equations like $38.4 = V \sin\alpha t - 4.9t^2$ were seen. Another incorrect approach was to assume that the ball was at its maximum height half way through the motion.

Some students attempted the question in an order that did not match the parts of the question. Typical of this was simply stating $V \cos\alpha$ in part (a) and then finding the value of 16 while doing part (b). It should be noted that if an answer of this type was required it would have been asked for in terms of V and α. (However please note that full marks were given to these students). Many also found the angle 39.1° while working on part (b). As a general principle, students should make sure that they give the answer that they intend for each part where they have written the part letter (eg (a)) on their script. In this question it did not cause issues for the marking, but in another case it could be more problematic.

Q4.

Students tended to do either very well or very badly on this question. This projectiles question, which did not contain a specified angle, seemed to confuse many students and some even tried to find an angle that they could use in some way. The problems were generally focussed on a confusion between the horizontal and vertical motion and the inability of students to relate the information given to these directions. For example in part (a), some students used 20 m to try to calculate the time, or in some cases the length of a line from the top of the cliff to the boat. In part (b), which required consideration of the horizontal motion an acceleration of 9.8 m s^{-2} was sometimes included. Similarly in part (c), the same type of confusion was present.
Also in Part (c), there were a few students who correctly found the two components of the velocity, but who did not then find the speed of the car.

Q5.

This question proved quite difficult, but candidates did tend to pick up marks as they progressed through their attempts. Many candidates gained 1 or 2 marks in part (a) for their attempt to find the angle. Some, however, tried to use the speed of 8 rather than the distances and scored zero.

In part (b), a reasonable number of candidates produced a quadratic equation, but often with incorrect signs, $8t$ instead of $8\sin\alpha t$ or the wrong distance. Once an error of this type was made, they began to lose marks, although those who showed how they solved their quadratic equation often gained the method mark. Even those with poor attempts at part (b) were able to gain some marks in part (c) by using $8\cos\alpha t$ and subtracting their answer from 10, although there were very few completely correct solutions.

Q6.

There were many correct answers to part (a) and the candidates were clearly helped by the presence of the printed answer. In most cases the arguments given were sound, but in others this was not the case. For example some candidates simply wrote statements such as 22.4 sin θ – 19.6 and never included an equals sign or a zero. This was regarded as an important omission.
There were also lots of good answers to part (b), but some basic errors, such as the omission of a negative sign with g or the use of sin(0875). In part (c) there were also some good answers. Some candidates did not realise that the time of flight could be found by doubling the 2 seconds that was given, and did quite a lot of work finding the time of flight. In some cases there were errors in the calculation of the distance. For example an incorrect application of $s = \frac{1}{2}(u + v)t$ was seen with statements such as $s = \frac{1}{2}(10.8 + 0) \times 4$.
The approaches to part (d) were many and varied. There were quite a number of good responses, but often candidates did not really get very far with this part of the question. The most common error was for candidates to use a distance, for example 19.6 – 5 = 14.6, with the initial speed of 22.4 sinθ.

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The approaches to part (d) were many and varied. There were quite a number of good responses, but often candidates did not really get very far with this part of the question. The most common error was to see candidates use a distance for example 19.6 – 5 = 14.6, with the initial speed of 22.4 sinθ.
In the final part, there was a tendency for the candidates to find the resultant of two

velocity components, as they did not realise that for the minimum speed the vertical component would be zero. In some case the correct final answer was obtained from expressions such as $\sqrt{10.8^2 + 0^2}$.

Q8.

The great majority of the candidates were able to answer both parts of this question correctly. However, a small number of candidates lost the last accuracy mark which was available for part (b) by not showing enough working. Centres should advise their candidates to show sufficient evidence of working when the answer to a question is given on the question paper.

Q9.

A large proportion of the candidates produced a quadratic equation, but there were several potential errors caused by setting up the equation in the wrong way. A few used the 1 or the 1.5 but not both, some used one with the incorrect sign and ended up with a 2.5 term instead of a 0.5 term, and some used 0.5 but with the wrong sign. Other methods that were seen included finding the time to move up and the time to move down and then adding them together. Some candidates lost one mark due to a lack of intermediate working and a failure to show how 1.30 was obtained from their quadratic equation. Again, showing an answer to more than three significant figures before quoting the final answer would have helped them.

Part (b) was usually done well with candidates using the printed answer from part (a).

In part (c), some candidates used the required approach to find the speed of the arrow, but many did not realise how to approach this part of the question. A few candidates made sign errors so that they obtained a vertical component of velocity that was too great ($12\sin 30° + 9.8 \times 1.3 = 18.74$). A few candidates only considered one component of the velocity. Those who had done well on part (c) were usually able to obtain the marks for part (d) too.

There were a lot of good answer to part (e), and in some cases this was the only place on the question where the candidate gained any marks.

Q10.

This question was more demanding and candidates sometimes confused the horizontal and vertical aspects of the motion, for example by including the 250 horizontal component of velocity in an equation for the vertical motion.
However, part (a) was generally done well. In part (b), the confusion between the horizontal and vertical components of the motion was most apparent,

with expressions like $s = 250 \times 4 + \frac{1}{2} \times 9.8 \times 4^2$ being seen. Candidates who took this approach in part (b) then often made little progress in the later parts of the question. Similar problems were evident in part (c) when candidates tried to calculate the vertical component of the velocity. In part (d), some candidates used distances rather than components of the velocity to calculate the angle. In some cases there was a mixture of a velocity and a distance!

Q11.

This question was answered very well by the great majority of the candidates. Part (b)(i) of the question was prone to errors for some candidates who did not take sufficient care in getting a correct simplified quadratic equation in tanθ and solving the equation to find the two possible values of θ Some of these candidates wrote 89.8° for a possible value of θ and did not seem to question or check the reasonableness of their answer. For part (b)(ii), almost all candidates chose their smaller angle for the answer, but many could not provide a convincing reason for their choice.

Q12.

The candidates seemed to be helped by the printed answer in part (a), but some obtained the printed result from the use of equations such as 5 = $\frac{1}{2}(-9.8)t^2$. There were also some candidates who did not justify all of their working. There were a number of candidates who found it difficult to work with the two components of the velocity in parts (b), (c) and (d).

A reasonably common error was to use 5 instead of 15 in part (b) and then 15 instead of 5 in part (c). Some candidates also stopped after obtaining the vertical component in part (c) and did not go on to find the speed. Those who did find the components correctly were usually able to go on to find the angle correctly.

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Q14.

Very many candidates were able to prove the result in part (a). Most considered the vertical component of the velocity and found the result easily. Some tried to find the time of flight first. In some cases these candidates halved their answer, but some did not do this or did not make it clear. A very small number of candidates used equations that they had learned for the time of flight or similar, but did not justify these results and so did not gain the marks. Candidates should not use learned formulae for the range etc in questions like this.

There were many different approaches to part (b)(i). Quite often candidates could produce a correct equation as a starting point, but failed to simplify it correctly. Often the candidates ended up with a mixture of sin α and sin^2 α terms. There were many minor arithmetic or algebraic errors. In part (b)(ii), many candidates used the incorrect time: their approach was based on a time of $\frac{3\sin\alpha}{2}$ instead of 3 sin α.

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Part (c) was often done well, but quite a few candidates incorrectly talked about the projectile having no mass.

Q16.

This question was answered very well and proved to be an easy source of marks for many candidates.

In part (a)(i), the candidates seemed comfortable with the kinematic equations of motion and the identity $\frac{1}{\cos^2\theta} = 1 + \tan^2\theta$. Very few candidates substituted 20 instead of –20 for y in answering part (a)(ii).

Some candidate lost an accuracy mark in part (b)(i) for premature rounding of the solutions of the equation for tan θ. Other candidates who wrote down inaccurate values for tan θ but used the accurate values to find the values of θ did not lose any accuracy mark. Centres should remind their candidates to give the **final** answer to questions to three significant figures, unless stated otherwise, as stated in the instructions on the front page of the question paper. In part (b)(ii), a small number of candidates gave the result to less than three significant figures.

Q17.

This question also provided more challenge for the candidates. In this question a number of candidates did assume that the acceleration due to gravity was positive.

In part (a), quite a number of candidates found the time of flight and then used half of this value to calculate the maximum height. There were however a number of cases where the candidates did not half their value for the time of flight and so produced solutions which did not relate to the maximum height.

Part (b) was much more demanding and relatively few correct solutions were seen. Those candidates who identified the required method were able to do well, but the majority made no attempt or produced working which could not lead to the required answer. Similarly part (c) was found to quite difficult, but some candidates did use the answer from part (b) to put together correct solutions for part (c) when they had been unable to do part (b).

Q18.

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In part (b), as might be expected some candidates had difficulty with the signs in their equations. Those who were able to form a reasonable quadratic often went on to solve this to find a value for the time. However quite a few candidates did not show both solutions of the quadratic. It is important for candidates to show both solutions and indicate which one they have selected. Some candidates did not show enough intermediate working to gain full marks in this "show that" type question.

A few candidates used a method which involved finding the time taken to move from the highest point to the point where it was caught and adding this to the time to reach the maximum height. This method was often used by candidates who had found a time in part (a).

In part (c) a large number of candidates simply found the vertical component of the velocity when the ball was caught and did not consider the horizontal component at all.

Q19.

Parts (a) and (b) were found to be fairly straightforward by candidates, but part (c) was much more demanding. The candidates often calculated the vertical component of the velocity in part (c), but were unable to use this correctly in further calculations. There was often confusion about how to calculate the speed and the direction of motion.

Q20.

The candidates found this question more demanding than the previous questions. In part (a), there were many good responses, but a number of candidates lost marks because they did not fully justify their final answer. When giving answers to questions like part (a), where an answer correct to three significant figures is required, candidates should be advised to write an answer to more than three significant figures before giving their final answer; otherwise examiners do not know whether the value has been calculated or copied from the question paper.

Part (b) was generally done well, sometimes using the answer given in part (a).

In part (c), there was quite a mixture of responses. A significant number of candidates thought that the distance would change. Of those who felt that the distance would not change, only some were able to justify this statement.

Part (d) was done well by some candidates, but one of the most common errors was to use a time of 3.13 seconds. A few candidates did not resolve the velocity and used 20 instead of 20 sin 50° in their calculations.

In part (e), candidates were simply expected to write down the velocity, but very few candidates did this. A significant number tried to calculate the velocity. In this case, there were often minor errors in the calculation of the speed. When stating the direction of the velocity, some candidates simply stated 50°, but did not specify that this was below the horizontal or draw a diagram to show this.

Q21.

Most candidates answered this question well. A small number of candidates used the given initial velocity components to find the initial speed and the tangent of the angle of projection. They then proceeded using the formula $y = x \tan \alpha - \frac{gx^2}{2v^2}(1 + \tan^2 \alpha)$.

For part (b), some candidates only found the two solutions of the quadratic equation in x and did not give the horizontal distance travelled by the particle whilst it is more than 1 metre above the plane. These candidates similarly gave two time limits for their answer to part (c). Many candidates failed to give the solutions to their quadratic equations to three significant figures.

Q22.

Some candidates found it very difficult to form a correct quadratic equation based on the vertical motion of the pellet. One of the main difficulties was dealing with the sign of the number 3.

Many candidates did part (b) well, and some who had been unable to find the time were able to use the printed answer to calculate the range. Parts (c) and (d) were found more challenging by candidates. While there were a few very good solutions, many candidates did not realise how to obtain the required answers, especially in part (d).

Q23.

Many candidates did well on this question, scoring a good number of marks. Part (a) was often answered well, but some candidates did include a number of spurious reasons. In part (b), there were many good responses. Some candidates did not show clearly how they obtained the equation which they went on to solve to find the time.

A large number of candidates found the time to the maximum height and doubled this value. Some of these candidates did not explain why this doubling had taken place and a 2 seemed to appear in their equations with no justification. Part (c) was generally done well. Some candidates gained full marks on part (c) without doing part (b). A few candidates had difficulty solving the inequalities to find the values of V.

Q24.

The candidates found this question more demanding, especially part (b). There were many good responses to part (a), but some candidates were not sure about how to go about finding the angle. Part (b) caused more difficulties. Some candidates were not able to form appropriate equations, and some of those who did made mistakes solving them. Some candidates found two times but did not find the difference between them.

Q25.

Parts (a) and (b) were generally done well. There were a few candidates who obtained the printed answer in part (a) from incorrect working. A popular approach was to find the time to the maximum height and to double this to find the time of flight. Part (b) was done very well, with some candidates who had not done part (a) using the printed answer to enable them to proceed.

Part (c) was more challenging and there were fewer correct completions. The main area of difficulty for this part of the question was the setting up of a quadratic equation. Many candidates included a 1 to allow for the change in height, but gave it the wrong sign. A number of candidates solved incorrect quadratic equations. A further issue concerned the solution of the quadratic. Some candidates showed both solutions and indicated that the solution required was the positive one; other candidates simply ignored the second solution without giving any justification for doing so.

In part (c), some candidates found the time by summing two times. For some of these students this approach worked well, but for others, especially when they involved the time to the highest point, there were quite a number of errors in their working.

Q26.

Parts (a), (b) and (c)(i) of this question were answered well by many candidates. However, some candidates lost one or more accuracy marks for parts (b) and (c)(i). To answer part (c)(i), most candidates evaluated $10 \cos 26.9°$ and $10 \cos 74.4°$ to arrive at the correct conclusions. A small number of candidates stated that the can will be knocked off the wall if $10 \cos \alpha > 8$, or $\cos \alpha > 0.8$, and then thought that this would imply $\alpha > 36.9°$.

Q27.

There were a fair number of good attempts at parts (a) and (b) of this question, but part (c) was found to be more demanding. In part (a), there were three main sources of error: introducing a sign error; using cos instead of sin; and not applying the quadratic equation formula correctly.

There were many good answers to part (b), with some candidates using the time given in part (a) when they had been unable to obtain it for themselves.

There were very few complete answers to part (c). Some candidates did not make any form of serious attempt at this part of the question. Some candidates only found one component of the velocity, but not the other; most often they found the vertical component, but not the horizontal component.

Q28.

Part (a) was usually done very well, but in some cases very badly. Many candidates produced good complete solutions. A few made arithmetic or algebraic errors that often resulted in the loss of the final accuracy mark. A few candidates found the time, but not the height. Part (b) was more demanding. Some candidates did not really make any progress with this part of the question. For those that got started, the most common errors were to include a sign error in the quadratic equation that they used to find the time; and to produce a correct quadratic equation, but to make an error in finding its solutions.

Some candidates did use a two-time approach, finding the time to go up to the maximum height and the time to come down to the tree. Many of these candidates were successful, but there were problems associated with getting the right distance to use to find the time down to the tree.

Q29.

This question caused problems for a number of candidates. The most common error, which created difficulties throughout the question, was to treat the ball as if it had an initial vertical component of velocity, often equal to 20 m s^{-1}. This would lead to equations of the form $2.45 = 4.9t^2 \pm 20t$. A few candidates who were unable to do part (a) were able to use the printed answer from part (a) to complete part (b) correctly, but some did include an initial vertical component of velocity again.

Part (c) of this question was more demanding. There were two main errors. The first was to work with the position of the ball when it hit the ground. These candidates would calculate an angle based on the two distances that they knew. A typical answer for these candidates was $\theta = \tan^{-1}\left(\frac{2.45}{14.14}\right)$. The other error was to only find one component of the ball's velocity.

Q30.

In part (a) the majority of the candidates gained full marks, although there were a small number of candidates with poor responses, such as "has no mass". Parts (b) and (c) were also done very well, but with some minor errors present. There were fewer correct responses to part (d).

In a number of cases the speed was given correctly and the angle of 40° was also stated, but no indication that it was below the horizontal was included. A few candidates worked out the horizontal and vertical components of the velocity when the arrow hit the ground and then calculated the speed and the direction. This was a very hard way to earn the marks, but it did work for some of these candidates. Generally, part (d) was either done well or very badly with candidates not knowing how to start.

Q31.

The responses to part (a) were very good. The vast majority of candidates were familiar with the kinematics equations for the motion of a projectile.

Many candidates answered part (b)(i) of the question correctly. However, some candidates misunderstood the axes and substituted –1 instead of 1 for y. Evidently, because the answer was given, some of these candidates were able to rectify their mistakes but others gave wrong answers.

The candidates understood and used the condition for real solutions for the quadratic equation to arrive at the relationship requested for part (b)(ii). For part (b)(iii), almost all candidates substituted 5 for R. However, some candidates were unable to proceed further and solve the quadratic equation in u^2 in order to find the minimum speed of projection. There were other candidates who attempted to use differentiation to find the minimum speed. These candidates tried to minimize the expression $u^4 - 20u^2 - 2500$ rather than finding the minimum value of u.

Q32.

This question proved difficult, and a number of candidates showed a lack of appreciation of the differences between the horizontal and vertical components of the motion. Solutions in part (a) sometimes lacked convincing reasoning. Algebra let some down in part (c), and part (d) proved very discriminating on the understanding of the motion.

Q33.

This question proved quite challenging, with the least successful candidates confusing the horizontal and vertical components of motion, and in some cases trying to incorporate expressions for the greatest height, horizontal range, and time of flight, thereby making the question harder. Part (a) was mostly done well, but some assumed vertical motion was involved. Part (b) was sometimes omitted, while a significant number did not realise that the calculation could be done in one stage and wasted time in carrying out a series of calculations often losing an accuracy mark due to rounding values within their working. In part (c)(i) common errors included finding the vertical component of velocity only, using '24' as the initial velocity with no component, and mixing horizontal/vertical motions or velocity/displacement in calculations. Those successful in part (c)(i) usually completed part (c)(ii).

Q34.

Part (a) The vast majority of candidates were familiar with the equations of motion for a projectile. They were able to eliminate t and arrive at the required equation of the trajectory.

Part (b) The algebraic manipulation and 'tidying up' of the quadratic equation in x and the subsequent task of solving it were prone to errors for some candidates. Some candidates who solved the equation correctly did not identify the required answer from their two solutions.

Part (c) Most candidates stated the two assumptions; no air resistance and the ball is a particle. A small minority of the candidates mistakenly stated that the ball was a particle without that mass.

Q35.

Part (a) of this question was done well by several candidates, while part (b) was found to be more demanding. In part (a), the candidates seemed to have been helped by the printed answer. In part (b), the major problem was setting up an appropriate equation to find the time of flight. Those who could do this generally went on to complete the question correctly. One issue that was evident was that a small number of candidates did not resolve the velocity into components, but simply worked with 30 wherever they needed a velocity.